

Helper Entities as Area Tiles

Approach & Impact Analysis — voice-orchestrator

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1. Problem

Tester report (Scott): a helper entity **Entry Door** assigned to the **Entry Area** drives an automation that pulses a Shelly relay's dry contact to open the door. The physical Shelly lives in a **Hidden Devices** area that is deliberately kept off dashboards. The app shows neither — the door has no visible control at all.

This is the standard Home Assistant pattern for dry-contact hardware: hide the raw relay, expose a clean virtual entity (helper) as the real control. HA's own area dashboard shows such helpers; ours does not.

2. Root Cause

The area views are built by `_expand_strategy_views` (app/controllers/voice_auth_controller.py:1463) from the device registry, which models the home as “tiles come from physical devices”. The helper is dropped twice:

#	Where	What happens
1	device_registry.py:252–256 (<code>_fetch_devices</code> step 3)	Entities are grouped by <code>device_id(entity)</code> ; any entity with no device — every helper, template entity, automation — is skipped as an “orphan”.
2	device_registry.py:270 + controller :1477–1480	Area is read only from the device (<code>area_name(device_id)</code>). A helper's area lives on the <i>entity</i> , which is never queried.

3. Proposed Design

Widen the model from “tiles come from devices that have an area” to “tiles come from entities that resolve to an area”, matching HA's own areas-strategy behavior. Four changes, bottom-up:

3.1 Registry — fetch entity-level areas in the existing template call. The per-entity template currently renders `device_id(e)`; change it to render a pair:

```
"<entity>": [device_id("<entity>") or "", area_name("<entity>") or ""]
```

HA's `area_name()` accepts an entity id and already resolves entity-level area with device-area fallback. Same single HTTP call, same 60 s cache — no new round-trips.

3.2 Registry — keep orphan entities that have an area. Device grouping stays exactly as-is. Alongside it, build a per-entity area map and expose a new cached method, e.g. `entity_areas(home_id)` -> `Dict[entity_id, area_name]`, plus the set of device-less (orphan) entities. No fake HADevice records — favorites, `/devices/discover` and `device_id_for_entity` all assume real devices.

3.3 Policy — which orphans qualify. Only controllable domains (registry `PRIMARY_DOMAINS`: light, lock, cover, climate, fan, media_player, switch, input_boolean). Device-less sensors, automations, scripts and zones stay off the boards. Agreed with Scott: helpers as tiles is the goal; widening to automations later is a

one-line set change.

3.4 Controller — group by entity area. `_expand_strategy_views` groups every entity by its *resolved* area (entity-level first, device fallback) and merges in qualifying orphans. Hidden/order slug logic is untouched — Hidden Devices stays hidden. Side benefit: an entity manually reassigned to a different area than its device now lands where HA shows it.

4. Impact Analysis

4.1 Component impact

Component	Impact	Notes
device_registry._fetch_devices	Modified	Template shape + parsing change; builds entity-area map alongside device list. Cache tuple widens from (ts, devices) to also carry the map — internal only, cleared on restart.
device_registry stale-cache fallback	Modified	Outage path must serve the cached entity-area map together with stale devices, so boards keep rendering through brief HA downtime.
_expand_strategy_views (boards)	Behavior change	The only client-visible change: helpers appear in their areas; reassigned entities move to their entity-level area.
HADevice dataclass	Unchanged	No schema change; all existing consumers keep working.
/devices/discover	Unchanged	Still device-based; device area still populated.
/items/search	Unchanged (optional later)	Bare entities return area: null today; the same map can fill it in — separable follow-up.
Favorites / device_id_for_entity	Unchanged	A favorited helper already degrades to “no device attached”; same behavior.
entity_metadata (tiles)	Unchanged	Keyed per entity_id; helpers get name/icon/state automatically. input_boolean is in CONTROLLABLE_DOMAINS (entity_icons.py:204), so the tile renders as tappable.
Dispatch (tile tap)	Unchanged	dispatch_direct defaults helpers to turn_on; entity existence pre-check already handles them. Toggling the helper fires Scott's automation.
Voice gating	Unchanged	Gate map is per entity_id; a helper can be gated like any entity.
Mobile app client	No change needed	New tiles arrive through the same views/entity_meta contract.

4.2 Risks and mitigations

Risk	Severity	Mitigation
Template payload roughly doubles (~50 KB at 400 entities); render cost grows with one area_name() per entity.	Low	Still one call, cached 60 s. If a 1000+ entity home appears, chunk the template — noted as future work, not needed now.
Template render error fails the whole fetch → boards 503.	Low	Same failure mode as today; 'or' guards keep area_name from raising, and the stale cache covers brief outages.
Entities move between area views (entity-level overrides now honored).	Medium	Intended — matches HA. Call it out to testers so moved tiles aren't reported as bugs.

Risk	Severity	Mitigation
Internal helpers testers assigned an area but never meant to show now appear.	Low	Area assignment on a helper is an explicit opt-in act in HA; HA's own dashboard shows them too. Removing the area hides them.
Registry unit tests mock the old template shape ({entity: device_id}).	Low	Update fixtures in tests/unit/test_home_unreachable.py; controller tests mock list_devices and additionally need the new entity_areas method stubbed.

5. Implementation Steps

Each step is independently shippable and ends at a checkpoint; code review before moving on.

Step	Work	Checkpoint
1. Registry data	Template renders [device_id, area] pairs; parse both; build entity-area map + orphan set; new entity_areas() method; cache and stale-fallback carry the map. Existing outputs byte-identical.	Unit tests green, incl. updated fixtures; no behavior change visible to any endpoint.
2. Board expansion	_expand_strategy_views groups by resolved entity area and merges controllable orphans; hidden/order logic untouched.	New tests: helper shows in its area; hidden area still excluded; reassigned entity moves; sensor orphan excluded.
3. Full suite + local run	Run the whole pytest suite in ./venv; fix fallout.	Suite green.
4. Deploy	Commit, push, droplet: git pull, docker-compose build app, up -d app (no migration needed — no DB change).	App healthy; /dashboards/config for a known home unchanged except expected additions.
5. Verify with Scott	Check scott_home /dashboards/config: Entry Door under Entry Area, Hidden Devices absent. Ask Scott to tap the tile.	Door opens; Scott confirms.
6. Optional follow-up	Fill area into /items/search entity rows from the same map.	Separate small PR.

6. Explicitly Out of Scope

- Showing device-less automations/scripts/scenes as tiles (policy set is one line away if wanted later).
- Door state feedback (the helper is stateless from the door's point of view; a contact sensor would be a separate feature).
- Any database or mobile-API contract change — there is none.